Aamodit Acharya

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Education

University of Waterloo — *Waterloo, ON*
Bachelor of Statistics with Computer Science Minor (Co-op) — *Sept 2021 - Dec 2026*

- **Courses:** Data Analysis, Stochastic Processes, Sampling and Experimental Design, Applied Linear Models, Forecasting

Work Experience

Wealthsimple — *Toronto, Ontario*
Data Science Intern — *Jan 2026 - Apr 2026*

- Increased mobile app engagement visibility by building a parallelized **Airflow** pipeline that checked users' **Braze** push-token registration, helping drive 25% higher engagement across marketing campaigns.
- Reduced manual attribute investigation by 75% by creating a daily **Braze** catalogue that used **Sourcegraph** to trace **dbt SQL** logic, reconciled **Hightouch** mappings, allowing for faster campaign troubleshooting.
- Reduced on-call debugging time by 30% by building a **Claude**-powered **Airflow** agent that maps DAG failure logs to **GitHub** source files via **Sourcegraph**, generating **Semgrep**-validated fixes for Slack and terminal review.

Mash — *Toronto, Ontario*
Data Science Intern — *Sept 2025 - Dec 2025*

- Improved answer relevance by 45% by building Mash's first **RAG** search pipeline using **Weaviate** hybrid search, **OpenAI embeddings**, and **Cohere** reranking to unify docs and Slack history.
- Expanded multimodal coverage by 50% by integrating **OpenAI Vision** into the Slack agent to extract text from screenshots and feed image-derived context into retrieval for Slack-based workflows.
- Reduced hallucinations by 30% by creating golden datasets from **Langfuse** traces and automating **Ragas** evals for faithfulness, recall, and relevance to validate all retrieval and prompt updates.

Statsig (Acquired by OpenAI) — *Bellevue, WA*
Data Science Intern — *May 2025 - Aug 2025*

- Cut compute time by 82% by replacing legacy count-distinct with **PySpark HLL++** sketches and pushing dedup upstream, generating mergeable unique-user estimates that made experimentation more reliable at Statsig's scale.
- Boosted segmentation reliability by orchestrating sketch pipelines with **Dagster** and enforcing table-level lineage with **Lakekeeper**, making backfills safer and reducing segmentation latency by 25% on **Iceberg** in **GCP**.
- Supported 5B daily events by building **Spark SQL** and **dbt** models that normalizes high-cardinality logs and prevent cartesian explosions, exposing a unified schema for fast reads in **Iceberg** and **BigQuery**.

TD Bank — *Montreal, QC*
Data Science Intern — *Jan 2024 - Apr 2024*

- Raised customer retention by 22% by building a cancellation-risk signal in **Impala SQL** on **Hadoop**, engineering real-time aggregated features that fed into downstream modeling and contract monitoring tools.
- Optimized processing time by 97% by rewriting legacy **R** scripts as **Impala SQL** queries, improving data refresh latency and stabilizing pipelines used for ML-driven pricing decisions.
- Informed pricing decisions by deploying a **Python** web-scraping job on **AWS EC2** with **Selenium**, **BeautifulSoup** and **Pandas** to collect timely market signals for underwriting and automated reporting workflows.

TD Bank — *Toronto, ON*
Data Science Intern — *May 2023 - Aug 2023*

- Decreased forecasting cycle time by 67% by building a **Python** framework with **Pandas** and **NumPy** on **AWS EC2** with **Docker** and **Kubernetes**, converting fragile single-VM scripts into reliable containerized services.
- Improved agent-closing-rate model performance by 25% by using **RidgeCV** and **linear regression** in **sk-learn**, replacing heuristic rules and strengthening predictive performance for sales funnel optimization.

Desjardins — *Toronto, ON*
Data Science Intern — *Sept 2022 - Dec 2022*

- Streamlined fraud detection accuracy by 30% by developing **SAS** code for the new-business-progress report to flag quote discrepancies from broker or customer inputs.
- Increased model accuracy by 10% across 10 risk factors by conducting a review of the New Brunswick segmentation model by ensuring alignment between **WTW-Radar** and **R** outputs.

Skills

Languages: Python, SQL, R, GraphQL, VBA, HTML/CSS
Technologies: NumPy, Pandas, Sk-learn, XGBoost, Pytorch, Git, Docker, Kubernetes, MySQL, Databricks, Azure, GCP, Snowflake